

CISC 3667 (TY11) Midterm Review Sheet

This review sheet covers the major concepts discussed in class so far.

1. What Is a Game?

Key Concepts

- **Game Design:** Creating a game's rules, mechanics, and experiences.
- **Iterative Design:** Repeatedly designing, testing, evaluating, and revising a game.
- **Lusory Attitude:** Willingly accepting a game's rules and limitations to participate in play.
- **Kevin Maroney:** A game is a form of play with goals and structure.
- **Jane McGonigal:** Games share four traits: goals, rules, feedback systems, and voluntary participation.
- **Jesse Schell:** A game is a problem-solving activity approached with a playful attitude.
- **Chris Crawford:** Distinguishes toys, challenges, puzzles, conflicts, competitions, and games based on goals, active opponents, and player interaction.
- **Bernard Suits:** Playing a game involves voluntarily attempting to overcome unnecessary obstacles.

2. What Is Fun?

Key Concepts

LeBlanc's Eight Kinds of Fun

- **Sensation:** Sensory enjoyment.
- **Fantasy:** Make-believe and imagined worlds.
- **Narrative:** Story and drama.
- **Challenge:** Overcoming obstacles.
- **Fellowship:** Social interaction.
- **Discovery:** Exploration and uncovering secrets.
- **Expression:** Creativity and self-expression.
- **Submission:** Relaxation and pastime.

Lazzaro's Four Keys 2 Fun

- **Hard Fun:** Challenge, mastery, and accomplishment.
- **Easy Fun:** Curiosity, exploration, and discovery.
- **Serious Fun:** Meaning, personal value, and emotional impact.
- **People Fun:** Social interaction and shared experiences.

Games can provide multiple kinds of fun simultaneously.

3. Player Types

Key Concepts

Bartle's Four Player Types

- **Achievers:** Enjoy accomplishments, points, progress, and completion.
- **Explorers:** Enjoy discovering game worlds and systems.
- **Socializers:** Enjoy interacting and forming relationships with other players.
- **Killers:** Enjoy competition, dominance, and acting on other players.

Players may demonstrate characteristics of multiple types.

4. The Elemental Tetrad & Game Mechanics

Key Concepts

The Elemental Tetrad consists of four interconnected elements:

- **Mechanics:** Rules, procedures, and actions.
- **Story:** Events and narrative.
- **Aesthetics:** The game's sensory presentation and experience.
- **Technology:** The tools and materials that make the game possible.

Objects, Attributes, and States

- **Object:** An entity in the game, such as a player or enemy.
- **Attribute:** A property of an object, such as health or speed.
- **State:** The current value of an attribute, such as 50 HP.

Basic Actions: Actions directly allowed by the rules.

Strategic Actions: Higher-level decisions that combine actions to achieve a goal.

5. Rules & Player Behavior

Key Concepts

Types of Rules

- **Operational:** Instructions describing how to play.
- **Foundational:** The underlying mathematical or formal structure.
- **Behavioral:** Implicit expectations about appropriate behavior and sportsmanship.
- **Written:** Rules documented for players.
- **House Rules:** Modifications created by players.

Player Behavior

- **Standard Player:** Follows the rules and plays as intended.
- **Dedicated Player:** Follows the rules while strongly pursuing mastery and victory.
- **Unsportsmanlike Player:** Follows formal rules but violates the spirit of play.
- **Cheat:** Breaks formal rules to gain an advantage.
- **Spoil-Sport:** Rejects or disrupts the game itself.

Degenerate Strategy: A strategy that is consistently effective and undermines meaningful choices by making other strategies less viable.

Exploit: Using an unintended feature or interaction in a game's system to gain an advantage.

6. Feedback Loops & Dynamic Difficulty Adjustment

Key Concepts

A feedback loop occurs when a system's output influences its future behavior.

- **Positive Feedback:** Amplifies an advantage or change, potentially causing a snowball effect. (Winning gives a player more resources, making future victories easier)
- **Negative Feedback:** Counteracts differences and moves the system toward equilibrium, often through rubberbanding or catch-up mechanics. (A racer who is behind receives stronger items)
- **Dynamic Difficulty Adjustment (DDA):** Automatically adjusts difficulty based on player performance. (A game provides additional health after repeated failures)

7. Randomness & Probability

Key Concepts

- **Randomness**: Events with unpredictable outcomes.
- **Probability**: The mathematical likelihood of an outcome.
- **Theoretical Probability**: The mathematically expected likelihood.
- **Practical Probability**: The frequency observed during actual trials.
- **Expected Value (EV)**: The average outcome expected over many occurrences.
- **Gambler's Fallacy**: Incorrectly believing that an outcome is more likely because it has not occurred recently. ("I've lost five times, so I'm due for a win.")
- **Lucky Streak Fallacy**: Incorrectly believing that an outcome is more likely to continue because it has occurred repeatedly. ("I've won five times in a row, so I'm more likely to win again.")
- Remember that when rolling two six-sided dice, different totals do not have equal probabilities.

Important Formulas

$$P(A) = \frac{\text{Favorable outcomes}}{\text{Total possible outcomes}}$$

***This formula assumes all possible outcomes are equally likely.**

$$P(\text{not}A) = 1 - P(A)$$

$$EV = \sum(\text{Outcome} \times \text{Probability})$$

Bayes' Theorem: A formula used to update the probability of an event or hypothesis based on new evidence. Bayes' theorem can help game designers reason about uncertain information and update predictions based on observed player behavior.

$$P(A|B) = [P(B|A) \times P(A)] / P(B)$$

Where:

- **P(A|B)**: Probability of A given B.
- **P(B|A)**: Probability of B given A.
- **P(A)**: Prior probability of A.
- **P(B)**: Probability of B occurring.

8. Game Balance

Key Concepts

Game Balance: Adjusting game elements to create the intended experience while preventing unintended dominant strategies.

Types of Balance

- **Symmetrical:** Players have identical or equivalent resources and abilities.
- **Asymmetrical:** Players have different abilities or resources but remain competitively viable.
- **Circular:** Each option has strengths and weaknesses against other options
 - Example: Rock-Paper-Scissors.

Challenge: Games should avoid being so easy that players become bored or so difficult that they become frustrated.

Meaningful Choice: A decision that significantly affects gameplay or outcomes.

Dominant Strategy: An option that is consistently better than alternatives, potentially reducing meaningful choices.

Triangularity: The relationship between risk, reward, and certainty.

9. JavaScript & Game Logic

Key Concepts

- **Game Logic:** Programming instructions that apply rules, process actions, and update the game state.
- **Variables:** Store information such as score, health, and position.
- **Objects:** Group related attributes and values.
- **Arrays:** Store ordered collections of values; indexing begins at 0.
- **Functions:** Group instructions that execute when called.
- **Conditional Statements:** Use if, else if, and else to apply rules based on conditions.

Important JavaScript Operators

Operator	Meaning
=	Assignment
===	Strict equality
!==	Strict inequality
>, <, >=, <=	Comparisons
&&	AND
	OR
!	NOT
++	Increase by 1
+=	Add and assign
-=	Subtract and assign

- **Events:** Actions such as clicks, form submissions, and key presses.
- **Event Listeners:** Execute code in response to events.
- **DOM (Document Object Model):** Allows JavaScript to access and modify HTML elements.
- **Game Loop:** Repeatedly processes input, updates game state, and renders the game.
- **Collision Detection:** Determines whether objects touch or overlap; the game's rules determine the consequence.

- **Delta Time:** Measures elapsed time between frames to support consistent movement.
- **Timing Functions:**
 - `setTimeout()` — Executes once after a delay.
 - `setInterval()` — Executes repeatedly at an interval.
 - `requestAnimationFrame()` — Schedules animation frames.

Sample Midterm Question Bank

The following sample questions are intended to help you prepare for the midterm. They cover the major topics discussed in class and demonstrate the types of questions you may encounter.

These questions are provided for practice and do not represent an exhaustive list of possible exam questions. You are responsible for understanding and applying the concepts covered in class.

Part A: Definitions & Short Response

Answer each question in approximately 1-3 sentences unless otherwise specified. When appropriate, provide an example to support your answer.

1. What four defining traits of games does Jane McGonigal identify?
2. Explain what is meant by lusory attitude. How does this concept relate to Bernard Suits' definition of a game?
3. What is iterative design, and why is playtesting important to this process?
4. Explain the difference between LeBlanc's Challenge and Discovery kinds of fun.
5. What is the difference between Lazzaro's Hard Fun and Easy Fun?
6. Explain the difference between Bartle's Achievers and Explorers.
7. Identify the four elements of the Elemental Tetrad and briefly describe each.
8. Explain the difference between an object, an attribute, and a state. Give an example from a game.
9. Explain the difference between a basic action and a strategic action.
10. What is the difference between operational rules and foundational rules?
11. Explain the difference between an unsportsmanlike player and a cheater.
12. What is a degenerate strategy, and why might it be a problem for game balance?
13. Explain the difference between positive and negative feedback loops.
14. What is Dynamic Difficulty Adjustment (DDA), and how might it improve the player experience?
15. Explain the difference between theoretical and practical probability.

16. What is the difference between the gambler's fallacy and the lucky streak fallacy?
17. What is the difference between symmetrical and asymmetrical balance?
18. Explain what makes a choice meaningful in game design.
19. What is triangularity, and how does it relate to risk and reward?
20. What is game state in JavaScript? Give two examples of variables that might represent a game's state.
21. Explain the purpose of `update()` and `draw()` in a JavaScript game loop.
22. What is collision detection, and how can it affect a game's state?

Part B: True or False

Determine whether each statement is true or false. For additional practice, explain why any false statement is incorrect.

1. According to Jane McGonigal, competition is one of the four defining traits of games.
2. A lusory attitude involves voluntarily accepting a game's rules and limitations.
3. LeBlanc's Eight Kinds of Fun are mutually exclusive, meaning a game can only provide one kind of fun at a time.
4. Bartle's Socializers primarily enjoy interacting and forming relationships with other players.
5. The four elements of the Elemental Tetrad can influence one another.
6. An attribute represents the current value of an object's property, while a state represents the property itself.
7. Behavioral rules involve expectations about appropriate player behavior and sportsmanship.
8. A player who uses an unusual but legal strategy is automatically cheating.
9. Positive feedback loops can cause an advantage to grow over time.
10. Negative feedback loops always make a game easier for every player.
11. Dynamic Difficulty Adjustment can respond to a player's performance.
12. When rolling two six-sided dice, all possible totals are equally likely.
13. If a fair coin lands on heads five times in a row, tails becomes more likely on the next flip.
14. Expected value represents the average outcome expected over many occurrences.
15. An asymmetrical game can still be balanced and fair.
16. A dominant strategy can reduce meaningful choices.
17. JavaScript arrays begin at index 1.
18. The JavaScript expression `score += 5` is equivalent to `score = score + 5`.
19. The logical OR operator (`||`) requires both conditions to be true.

- 20. The draw() function is generally responsible for changing a game's rules and updating its state.
- 21. A JavaScript object can store multiple attributes of a game character.
- 22. requestAnimationFrame() can be used to repeatedly schedule frames in a JavaScript game.

Part C: Multiple Choice

Choose the best answer for each question.

1. Which of the following best represents Jesse Schell's definition of a game?
 - A. A form of play with goals and structure.
 - B. A voluntary attempt to overcome unnecessary obstacles.
 - C. A problem-solving activity approached with a playful attitude.
 - D. An activity defined by goals, rules, feedback systems, and voluntary participation.
2. A child uses a collection of building blocks to construct towers and experiment with different structures. There is no specific goal, scoring system, or opponent. According to Chris Crawford's classification, this activity is best described as a:
 - A. Puzzle
 - B. Toy
 - C. Challenge
 - D. Game
3. Two players participate in a timed puzzle-solving contest. Each solves an identical puzzle independently, and neither can interfere with the other's progress. The player who finishes first wins. According to Crawford's classification, this is best described as:
 - A. A puzzle
 - B. A conflict
 - C. A competition
 - D. A game

4. A player enjoys customizing their character, designing a house, and creating outfits. They also enjoy exploring the game's world to discover hidden furniture and decorations. Which two of LeBlanc's kinds of fun are most directly represented?

- A. Expression and Discovery
- B. Fantasy and Challenge
- C. Fellowship and Submission
- D. Narrative and Sensation

5. Which of the following best demonstrates the difference between a basic action and a strategic action?

- A. Pressing a button to jump is a strategic action, while deciding when to jump is a basic action.
- B. Moving a chess piece is a basic action, while sacrificing a piece to gain a positional advantage is a strategic action.
- C. Basic actions are always simple, while strategic actions must involve multiple players.
- D. Basic actions are defined by the game's story, while strategic actions are defined by its technology.

6. A board game's rules state that players must roll a six-sided die and move that many spaces. A group decides that players must instead roll two dice and move according to their sum. Which statement best describes this change?

- A. The group has changed a behavioral rule because players are expected to follow the new procedure.
- B. The group has created a house rule that modifies the game's operational rules.
- C. The group has changed only the game's aesthetics because the objective remains the same.
- D. The group has created a foundational rule without affecting how the game is played.

7. A player discovers a legal strategy that makes it extremely difficult for opponents to respond. Although the strategy does not violate the game's formal rules, the community considers repeated use of it unfair. Which statement is most accurate?

- A. The player is automatically cheating because the strategy is unfair.
- B. The player is a spoil-sport because they are attempting to win.
- C. The strategy may be degenerate, and its use may raise concerns about behavioral rules without necessarily being cheating.
- D. The strategy cannot be considered problematic because it is permitted by the formal rules.

8. In a strategy game, capturing territory provides additional resources. These resources allow players to build stronger armies, making it easier to capture more territory. Which statement best describes this system?

- A. It is a negative feedback loop because capturing territory changes the game's state.
- B. It is a positive feedback loop because an initial advantage can create further advantages.
- C. It is DDA because the game automatically changes difficulty.
- D. It is circular balance because players have multiple strategic options.

9. A racing game gives stronger items to players who are behind. However, a player who is far ahead continues to receive powerful items because of a programming error. Which statement best describes the intended system and the possible effect of the error?

- A. The intended system uses positive feedback, and the error creates negative feedback.
- B. The intended system uses negative feedback, while the error may reinforce the leading player's advantage.
- C. Both systems are examples of negative feedback because items are distributed randomly.
- D. Both systems are examples of DDA because the game changes item distribution.

10. A player rolls a fair six-sided die 120 times and records 26 rolls of a 6. Based on these results, which statement is most accurate?

- A. Rolling a 6 occurred approximately 5 percentage points less frequently than theoretically expected, suggesting the die may be biased.
- B. Rolling a 6 occurred approximately 5 percentage points more frequently than theoretically expected, but this does not change the probability of rolling a 6 on the next roll.
- C. Rolling a 6 occurred approximately 5 percentage points more frequently than theoretically expected, suggesting this pattern is likely to continue.
- D. Rolling a 6 occurred approximately 5 percentage points less frequently than theoretically expected, but the difference should disappear after enough additional rolls.

11. A fighting game has two characters. Character A is fast but deals low damage, while Character B is slow but deals high damage. Both are designed to have reasonable chances of winning. Which statement is most accurate?

- A. The game demonstrates symmetrical balance because both characters can win.
- B. The game demonstrates asymmetrical balance because different abilities can remain competitively viable.
- C. The game demonstrates circular balance because one character must always defeat the other.
- D. The game is necessarily unbalanced because the characters have different abilities.

12. Consider the following JavaScript code:

```
let score = 8;

let health = 20;

let gameOver = false;

if (score >= 10 || health <= 20) {

    score += 5;

}

if (score >= 10 && health > 0) {

    gameOver = true;

}
```

What are the final values of score and gameOver?

- A. score = 8, gameOver = false
- B. score = 13, gameOver = false
- C. score = 13, gameOver = true
- D. score = 8, gameOver = true

Part D: Probability & Game Balance Calculations

Show your work and explain your reasoning where appropriate.

Question 1: Basic Probability

A player rolls a fair six-sided die.

- What is the probability of rolling a number greater than 4?
- What is the probability of NOT rolling a number greater than 4?
- Express both answers as fractions.

Question 2: Multiple Dice

A player rolls two fair six-sided dice.

- How many possible ordered outcomes are there?
- What is the probability of rolling a total of 7?
- What is the probability of rolling a total of 12?
- Explain why these probabilities differ.

Question 3: Practical Probability

A player rolls a six-sided die 80 times and rolls a 6 on 12 occasions.

- What is the practical probability of rolling a 6?
- What is the theoretical probability?
- Explain why the two values might differ.

Question 4: Expected Value

A treasure chest has the following possible rewards:

Reward	Probability
2 coins	50%
5 coins	30%
10 coins	20%

- Calculate the expected value of opening the chest.
- Explain what the expected value represents.
- Does the expected value guarantee that the player will receive that number of coins from one chest? Explain.

Question 5: Risk, Reward & Triangularity

A player can choose between two attacks:

- Safe Attack: Deals 6 damage with 100% probability.
- Risky Attack: Deals 18 damage with 40% probability and 0 damage otherwise.

- Calculate the expected damage of each attack.
- Which attack has the higher expected value?
- Explain why a player might still choose the attack with the lower expected value.
- How does this example relate to triangularity?

Question 6: Bayes' Theorem

A game has two types of treasure chests:

- 40% of chests are rare.
- 60% of chests are common.
- A rare chest has an 80% chance of containing a gem.
- A common chest has a 20% chance of containing a gem.

A player opens a chest and finds a gem.

a. What is $P(\text{Rare})$?

b. What is $P(\text{Gem}|\text{Rare})$?

c. Calculate the overall probability of finding a gem, $P(\text{Gem})$.

d. Use Bayes' theorem to calculate $P(\text{Rare}|\text{Gem})$, the probability that the chest was rare given that it contained a gem.

Part E: JavaScript Code Tracing

Question 1

Consider the following JavaScript code for a Tic-Tac-Toe game.

Carefully follow the code in the order it executes. Track changes to variables and game state, and answer the following questions:

```
1  let board = [  
2    "X", "O", "",  
3    "", "X", "",  
4    "O", "", ""  
5  ];  
6  
7  let winningConditions = [  
8    [0, 1, 2],  
9    [3, 4, 5],  
10   [6, 7, 8],  
11   [0, 3, 6],  
12   [1, 4, 7],  
13   [2, 5, 8],  
14   [0, 4, 8],  
15   [2, 4, 6]  
16  ];  
17  
18  let currentPlayer = "X";  
19  let winner = "";  
20  
21  function checkWinner() {  
22    for (let condition of winningConditions) {  
23      let a = condition[0];  
24      let b = condition[1];  
25      let c = condition[2];  
26  
27      if (  
28        board[a] !== "" &&  
29        board[a] === board[b] &&  
30        board[b] === board[c]  
31      ) {  
32        winner = board[a];  
33        return;  
34      }  
35    }  
36  }  
37  
38  function makeMove(position) {  
39    if (board[position] !== "" || winner !== "") {  
40      return;  
41    }  
42  
43    board[position] = currentPlayer;  
44    checkWinner();  
45  
46    if (winner === "") {  
47      currentPlayer = currentPlayer === "X" ? "O" : "X";  
48    }  
49  }  
50  
51  makeMove(2);  
52  makeMove(3);  
53  makeMove(8);  
54  makeMove(5);  
55  makeMove(7);
```

Questions:

- a. What are the final values stored in **board[2]**, **board[3]**, **board[5]**, **board[7]**, and **board[8]**?
- b. What is the final value of **currentPlayer**?
- c. What is the final value of **winner**? Which player, if any, wins the game?
- d. Which **winning condition**, if any, is satisfied? Identify the corresponding indices from the winningConditions array.

Part F: JavaScript Code Writing

Write short JavaScript code snippets that satisfy the following requirements. You do not need to write a complete HTML page or game.

Question 1: Score & Winning Condition

Write a function called `addPoints(points)` that:

- Adds points to the player's score.
- Checks whether the score is at least 20.
- Sets `gameWon` to true if the condition is met.

Assume score starts at 0 and `gameWon` starts as false.

Question 2: Player Movement

Given the following object:

```
const player = {  
  x: 50,  
  speed: 5  
};
```

Write JavaScript code that:

- Moves the player to the right using their speed.
- Ensures the player cannot move beyond the right boundary of the game area, located at `x = 100`.

Question 3: Collecting Items

A game uses the following variables:

```
let score = 0;  
let itemsCollected = 0;
```

Write a function called `collectItem()` that:

- Increases `itemsCollected` by 1.
- Increases `score` by 10.
- Awards an additional 20 points when `itemsCollected` reaches exactly 5.

Question 4: Health & Game Over

Given:

```
let health = 30;  
let gameOver = false;
```

Write a function called `takeDamage(amount)` that:

- Subtracts `amount` from `health`.
- Sets `gameOver` to `true` if `health` is less than or equal to 0.